# EdgeReliant-AI

## Machine Learning-Based Multi-Sensor Fault Classification and Engineering Decision Support

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| VIEW GITHUB REPOSITORY | PROJECT RESULTS | SYSTEM ARCHITECTURE |
|---|---|---|

| PROJECT STATUS | TECHNOLOGY STACK |
|---|---|
| **Functional Machine-Learning Prototype** | Python · Pandas · NumPy · Scikit-learn · Matplotlib · Joblib |

**Repository**

github.com/HAFSAH-SAEED/EdgeReliant-AI

***From Sensor Data to Engineering Decisions***

Sensor Data → ML Classification → Confidence → Severity → Action → Verification

Predictive-maintenance prototype using simulated multi-sensor data

## Abstract

Modern engineering systems increasingly rely on sensor measurements to monitor operating conditions and identify potential faults before they lead to system degradation or failure. Machine learning provides a potential approach for extracting patterns from multi-sensor measurements and associating those patterns with predefined machine conditions.

EdgeReliant-AI is a functional machine-learning and predictive-maintenance prototype developed to explore multiclass fault classification from simulated multi-sensor measurements. The system uses temperature, vibration, pressure, voltage, and current measurements to represent five machine conditions: Normal, Overheating, High Vibration, Pressure Anomaly, and Electrical Instability.

The project follows an engineering-oriented workflow in which simulated sensor data is generated and analyzed, Logistic Regression and Random Forest classifiers are trained and compared, and the resulting prediction is connected to confidence estimation, fault severity, recommended action, and verification guidance.

The final dataset contains 5,000 observations distributed equally among the five fault classes. Logistic Regression achieved an accuracy of 90.60%, macro precision of 91.82%, macro recall of 90.60%, and macro F1-score of 90.89%. Random Forest achieved an accuracy of 89.50%, macro precision of 90.41%, macro recall of 89.50%, and macro F1-score of 89.74%. Logistic Regression was therefore selected as the final model.

The reported results demonstrate that the simulated sensor space contains class-dependent structure that can be learned by supervised classifiers. However, these results describe performance on simulated data only and should not be interpreted as evidence of industrial reliability. The project is therefore best understood as an engineering machine-learning prototype that provides a foundation for more rigorous future experiments.

## 1. Introduction

### 1.1 Background

Engineering systems frequently depend on continuous measurements of physical variables such as temperature, vibration, pressure, voltage, and current. Changes in these measurements can provide information about the operating condition of a machine or system.

Traditional condition monitoring approaches may rely on fixed thresholds or manually defined diagnostic rules. While threshold-based methods can be effective for clearly defined conditions, engineering systems often produce interacting measurements across multiple sensors. Machine-learning methods provide an alternative approach in which patterns in historical measurements can be used to classify system conditions.

Predictive maintenance is one application of this concept. Rather than responding only after a failure occurs, an intelligent monitoring system can attempt to identify abnormal operating conditions and provide information that supports maintenance decisions.

EdgeReliant-AI was developed as a practical exploration of this idea, combining simulated sensor generation, machine-learning classification, exploratory analysis, and a downstream engineering decision layer. The project is intentionally positioned as a prototype , it does not claim to represent a validated industrial predictive-maintenance system, and the current implementation uses simulated rather than physical sensor data.

## 2. Problem Definition

The central problem addressed by EdgeReliant-AI is the classification of machine operating conditions using multiple sensor measurements. For each observation, the system receives a feature vector X = [Temperature, Vibration, Pressure, Voltage, Current]. The objective is to estimate a machine condition y, where y belongs to one of five predefined classes:

- Normal
- Overheating
- High Vibration
- Pressure Anomaly
- Electrical Instability

The project therefore represents a multiclass classification problem. The broader engineering objective extends the prediction into a decision-support sequence:

***Sensor Data → Validation → Sensor Analysis → ML Classification → Confidence → Severity → Recommended Action → Verification***

## 3. Objectives

- To construct a simulated multi-sensor machine dataset.
- To represent several predefined machine operating and fault conditions.
- To examine the resulting sensor behaviour through exploratory analysis.
- To investigate whether machine-learning models could classify the simulated conditions.
- To compare Logistic Regression and Random Forest classification performance.
- To evaluate the resulting models using multiple classification metrics.
- To investigate the relationship between different sensor variables.
- To connect model predictions to a prototype engineering decision-support layer.
- To provide recommended actions and verification guidance associated with predicted faults.
- To establish a foundation that could subsequently be extended toward more rigorous research experiments.

**The project was not designed as a formal industrial validation study**. Its primary contribution at the current stage is the integration of machine learning with a simulated engineering monitoring workflow.

## 4. System Architecture

### 4.1 Sensor Data Generation

The system begins with simulated measurements representing machine behaviour under predefined conditions, using five variables: temperature, vibration, pressure, voltage, and current. The dataset was developed iteratively as additional fault classes and sensor variables were introduced.

### 4.2 Data Validation

The final dataset was checked during development for missing values and duplicate rows; the reported result was zero missing values and zero duplicate rows.

The final dataset has shape 5,000 × 7: temperature, vibration, pressure, voltage, current, fault, and timestamp. It contains exactly 1,000 observations per class.

```
Dataset shape:
(5000, 7)

Fault distribution:
fault
Normal                  1000
Overheating             1000
High Vibration          1000
Pressure Anomaly        1000
Electrical Instability  1000
Name: count, dtype: int64

First five rows:
                   timestamp  temperature  vibration  pressure  voltage  current   fault
0 2026-08-13 06:39:44.383986        45.99      0.196    101.95    12.23     1.77  Normal
1 2026-08-13 06:39:45.383986        44.53      0.247    102.07    11.93     1.87  Normal
2 2026-08-13 06:39:46.383986        44.07      0.186    101.54    11.71     1.59  Normal
3 2026-08-13 06:39:47.383986        43.88      0.170    101.61    11.86     1.63  Normal
4 2026-08-13 06:39:48.383986        47.93      0.193    101.37    11.79     1.73  Normal
```

***Figure A. Terminal output confirming final dataset shape (5000, 7) and balanced class distribution (1000 per class).***

### 4.3 Computational Stages

The project is organized into separate computational stages for model training, prediction, multi-fault testing, and model comparison. These stages were executed repeatedly during development and produce saved outputs for evaluation.

### 4.4 Model Artifact

The final repository uses models/final_model.pkl as the final trained model artifact. An earlier experimental artifact named trained_model.pkl was used during development and is excluded from version control through the project's .gitignore.

## 5. Simulated Sensor Environment

## 5.1 Temperature

Temperature was used to represent thermal behaviour under different machine conditions. The simulated visualizations show a comparatively stable lower-temperature region for Normal operation and a substantially elevated temperature region for Overheating.

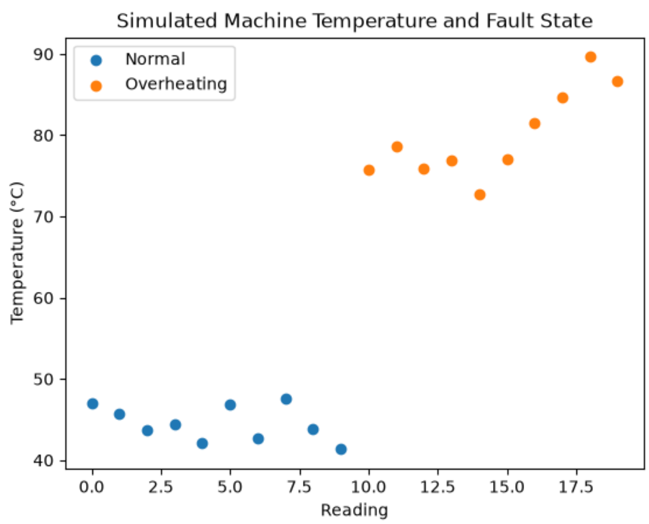


*Figure 1. Initial temperature fault-state simulation — Normal vs. Overheating.*

## 5.2 Vibration

The Normal and Overheating conditions occupy relatively low vibration ranges, while the High Vibration condition occupies a substantially higher, well-separated range in the simulated data.

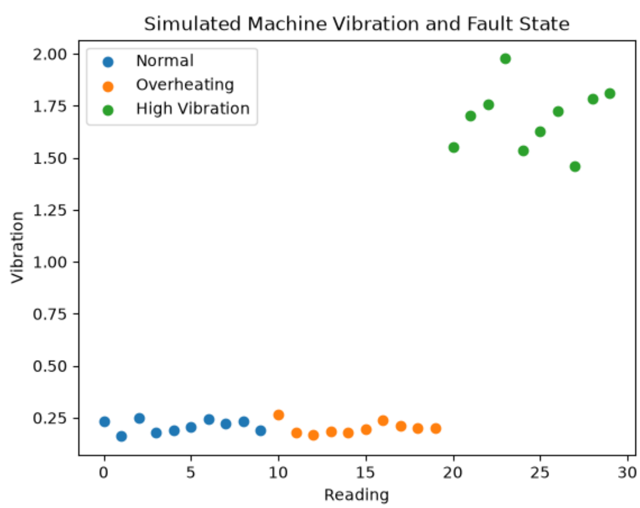


*Figure 4. Three-state vibration simulation, showing the strong vibration signature associated with the High Vibration condition.*

## 5.3 Pressure

Pressure remains relatively concentrated around the nominal operating region for several classes. The Pressure Anomaly condition is represented by a pronounced downward pressure trajectory, creating a distinguishable sensor signature.

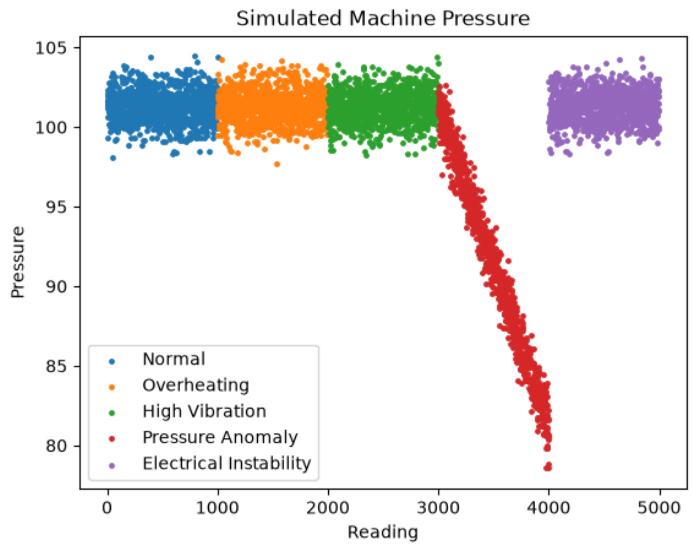


*Figure 7. Five-state pressure simulation showing the pressure signature associated with Pressure Anomaly.*

## 5.4 Voltage and Current

Voltage and current represent electrical behaviour, with Electrical Instability distinguished by changes in these two variables. The correlation analysis shows the strongest pairwise relationship in the dataset is between voltage and current, at approximately -0.67.

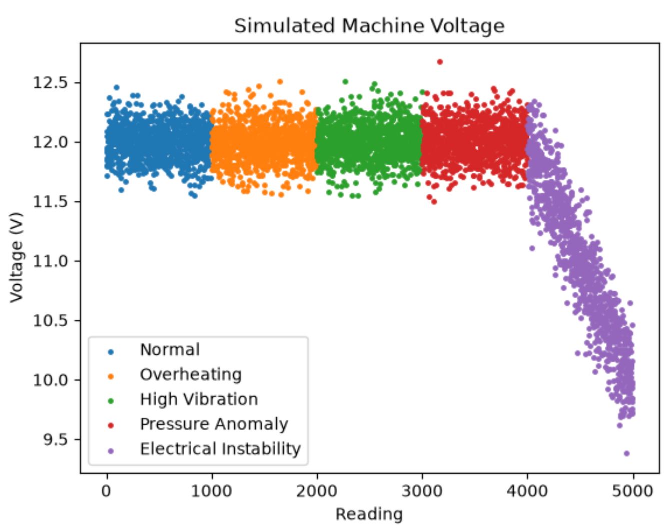

*Figure 8. Five-state voltage simulation showing electrical behaviour across the five machine conditions.*

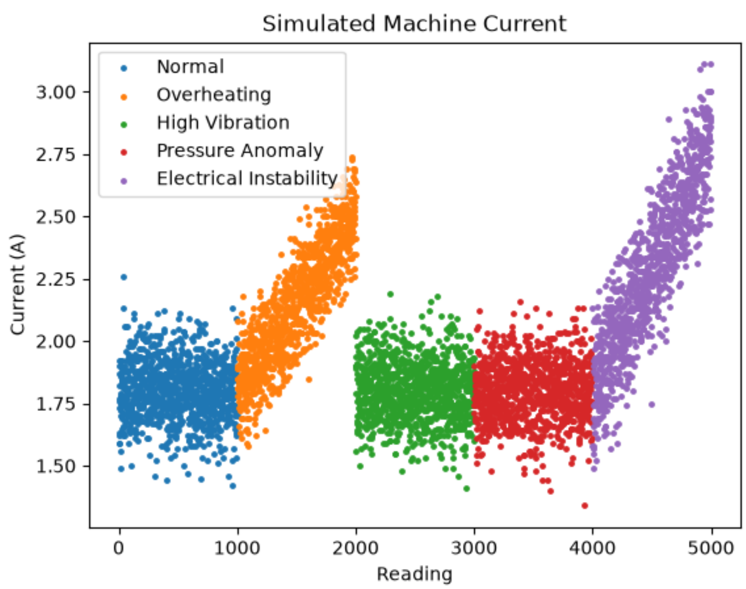


*Figure 9. Five-state current simulation.*

## 6. Exploratory Data Analysis

Exploratory analysis examined the sensor variables and fault states through class-balance inspection, sensor visualizations, fault-specific behaviour, and a dedicated correlation matrix.

- Overheating is strongly associated with elevated temperature.
- High Vibration is strongly associated with elevated vibration.
- Pressure Anomaly is associated with a pressure decrease.
- Electrical Instability is associated with changes in voltage and current.
- Normal operation occupies comparatively stable operating regions.

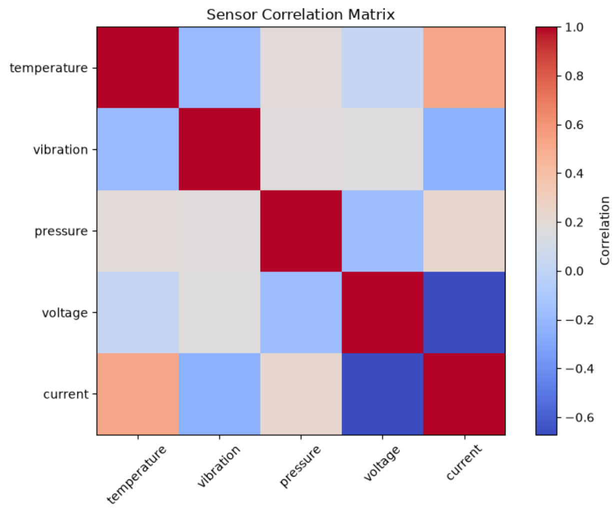


*Figure 10. Sensor correlation matrix — voltage and current show the strongest pairwise relationship (≈ -0.67).*

These relationships provide the statistical structure from which the classification models learn and help explain why several of the simulated fault classes are relatively separable.

## 7. Dataset Construction

The final dataset contains 5,000 observations.

| Property | Value | Status |
|---|---|---|
| Total observations | 5,000 | Confirmed |
| Sensor features | 5 | Confirmed |
| Fault classes | 5 | Confirmed |
| Samples per class | 1,000 | Confirmed |
| Missing values | 0 | Stated |
| Duplicate rows | 0 | Stated |

The dataset was developed progressively from earlier versions with fewer fault classes and sensor variables. The final version contains five classes and five sensor features, providing the basis for the final multiclass classification experiments.

```
Dataset shape:
(4000, 5)

Fault distribution:
fault
Normal              1000
Overheating         1000
High Vibration      1000
Pressure Anomaly    1000
Name: count, dtype: int64

First five rows:
                   timestamp  temperature  vibration  pressure   fault
0 2026-08-13 06:35:45.730322        45.99      0.196    101.95  Normal
1 2026-08-13 06:35:46.730322        48.05      0.193    101.07  Normal
2 2026-08-13 06:35:47.730322        48.16      0.223    100.83  Normal
3 2026-08-13 06:35:48.730322        46.09      0.186    100.83  Normal
4 2026-08-13 06:35:49.730322        45.48      0.143     99.58  Normal

Dataset saved to data/sensor_data.csv
```

***Figure B. Terminal output from an earlier development stage (4,000 rows, four fault classes) showing the dataset's incremental growth.***

## 8. Machine Learning Methodology

Two supervised classification algorithms were evaluated: Logistic Regression and Random Forest, trained on the five sensor measurements as input features.

The final evaluation uses an 80/20 train/test split, corresponding to 4,000 training samples and 1,000 testing samples.

## 9. Logistic Regression

Logistic Regression estimates a probability for each possible class and assigns the observation to the class with the highest estimated probability. These class probabilities are also used by the prototype to report prediction confidence.

## 10. Random Forest

Random Forest was evaluated as a nonlinear ensemble-based alternative to Logistic Regression. Its feature-importance estimates provide an additional view of which simulated sensor variables contribute most strongly to classification.

| Sensor | Importance |
|---|---|
| Vibration | 0.263698 |
| Pressure | 0.240904 |
| Temperature | 0.208652 |
| Voltage | 0.188370 |
| Current | 0.098376 |

Vibration has the largest reported Random Forest feature importance, followed by pressure and temperature. This does not mean vibration is universally the most important sensor for machine fault diagnosis in general — it means that, within this particular simulated dataset and model, vibration produced the largest measured contribution.

```
results_random_forest.txt
TRAINING SAMPLES: 4000
TESTING SAMPLES: 1000

MODEL ACCURACY
0.8950

CLASSIFICATION REPORT
                      precision    recall  f1-score   support

Electrical Instability     0.95      0.84      0.89       200
        High Vibration     0.97      0.94      0.96       200
                Normal     0.72      0.86      0.79       200
           Overheating     0.92      0.93      0.92       200
      Pressure Anomaly     0.96      0.90      0.93       200

              accuracy                         0.90      1000
             macro avg     0.90      0.90      0.90      1000
          weighted avg     0.90      0.90      0.90      1000


CONFUSION MATRIX
[[168   2  25   5   0]
 [  1 189  10   0   0]
 [  6   3 173  12   6]
 [  1   1  12 185   1]
 [  1   0  19   0 180]]

FEATURE IMPORTANCE
vibration      0.263698
pressure       0.240904
temperature    0.208652
voltage        0.188370
current        0.098376
dtype: float64
```

*Figure C. Random Forest results file showing training/testing sample counts, classification report, confusion matrix, and feature importance values.*

## 11. Experimental Evaluation

The models were evaluated using accuracy, macro precision, macro recall, macro F1-score, and confusion matrices. Macro averaging gives equal weight to each fault class, appropriate here since the dataset is class-balanced.

## 12. Model Comparison

| Model | Accuracy | Macro Precision | Macro Recall | Macro F1 |
|---|---|---|---|---|
| Logistic Regression | 90.60% | 91.82% | 90.60% | 90.89% |
| Random Forest | 89.50% | 90.41% | 89.50% | 89.74% |

Logistic Regression outperformed Random Forest on all four reported metrics, by modest but consistent margins. The simpler model was selected because it achieved the stronger measured performance rather than being chosen by default.

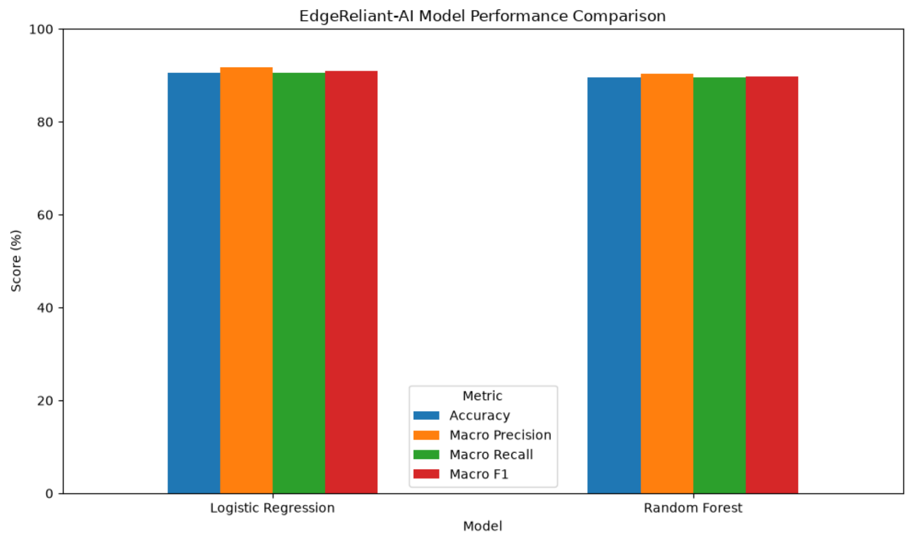


*Figure 13. Model performance comparison across accuracy, macro precision, macro recall, and macro F1 for both models.*

## 13. Confusion Matrix Analysis

The Logistic Regression confusion matrix provides class-level detail beyond aggregate accuracy.

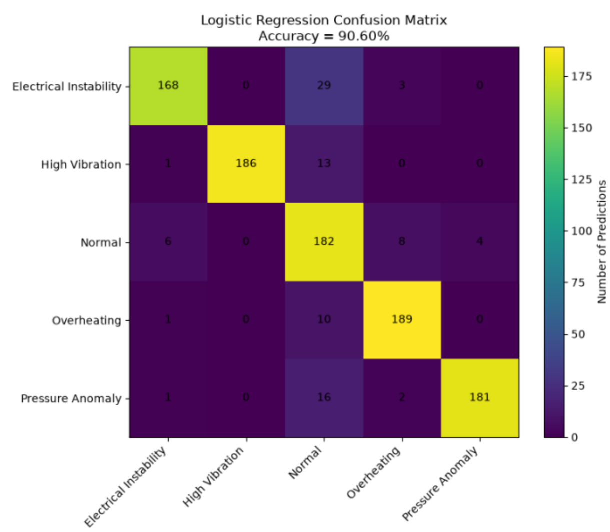


*Figure 11. Logistic Regression confusion matrix (Accuracy = 90.60%).*

High Vibration (186/200) and Overheating (189/200) are the most cleanly separated classes. Electrical Instability has the weakest diagonal performance (168/200), with 29 samples misclassified as Normal. The Normal class itself contributes to several misclassification patterns across the matrix , a detail lost if only aggregate accuracy is reported.

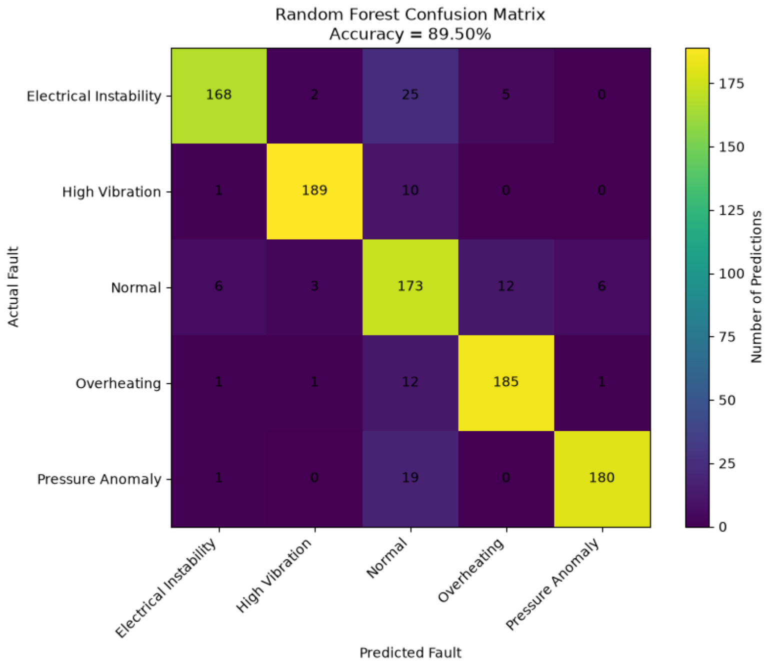


*Figure 12. Random Forest confusion matrix (Accuracy = 89.50%), shown for comparison.*

## 14. Error Structure and Class Separability

The confusion matrix shows the most significant ambiguity involves the Normal condition , Electrical Instability is sometimes classified as Normal, and Normal itself is occasionally confused with Pressure Anomaly, Overheating, and Electrical Instability. This is consistent with the lower (63.87%) confidence observed for the Normal case in multi-fault testing (Section 15), and raises a genuine, evidence-motivated question for future work: which sensor measurements are responsible for distinguishing normal operation from less strongly expressed fault conditions?

## 15. Multi-Fault Prediction Testing

The multi-fault testing script was used to evaluate representative inputs across all five fault classes.

```
EDGE RELIANT AI - MULTI-FAULT TEST
============================================================

RESULTS
------------------------------------------------------------
              Test Case        Predicted Fault  Confidence
                 Normal                 Normal       63.87
            Overheating            Overheating      100.00
         High Vibration         High Vibration      100.00
       Pressure Anomaly       Pressure Anomaly      100.00
 Electrical Instability Electrical Instability      100.00

RESULTS SAVED
multi_fault_test_results.csv
PS C:\Users\Umer work\Desktop\EdgeReliant-AI> python model_comparison.py

EDGE RELIANT AI - MODEL COMPARISON
======================================================================
              Model  Accuracy  Macro Precision  Macro Recall  Macro F1
Logistic Regression     0.906         0.918235         0.906  0.908885
      Random Forest     0.895         0.904128         0.895  0.897399

RESULTS SAVED
model_comparison_results.csv
```

*Figure 14. Multi-fault prediction terminal output and model comparison table, generated by test_predictions.py and model_comparison.py.*

| Test Case | Predicted Fault | Confidence |
|---|---|---|
| Normal | Normal | 63.87% |
| Overheating | Overheating | 100.00% |
| High Vibration | High Vibration | 100.00% |
| Pressure Anomaly | Pressure Anomaly | 100.00% |
| Electrical Instability | Electrical Instability | 100.00% |

All five representative cases were correctly classified. **This test should not be treated as an independent held-out validation set unless the test cases were generated separately from the training data. The lower confidence for Normal is a genuine and informative result and should be reported as-is.**

## 16. Example End-to-End Prediction

An end-to-end prediction run demonstrates the complete implemented output from sensor input through fault classification, confidence, severity, recommended action, and verification guidance.

```
EDGE RELIANT AI - FAULT PREDICTION
----------------------------------
Enter temperature: 75
Enter vibration: 0.20
Enter pressure: 101
Enter voltage: 12
Enter current: 2.4

PREDICTION
----------------------------------
Predicted Fault: Overheating
Confidence: 100.00%
Severity: HIGH

Recommended Action:
Inspect the cooling system and reduce machine load if necessary.

Verification:
Check temperature again after corrective action and confirm that it returns toward the normal operating range.
PS C:\Users\Umer work\Desktop\EdgeReliant-AI>
```

*Figure D. End-to-end prediction: sensor input through to fault, confidence, severity, recommended action, and verification guidance.*

| Sensor | Value |
|---|---|
| Temperature | 75 |
| Vibration | 0.20 |
| Pressure | 101 |
| Voltage | 12 |
| Current | 2.4 |

**Output:** Predicted Fault: Overheating | Confidence: 100.00% | Severity: HIGH

Recommended Action: Inspect the cooling system and reduce machine load if necessary.

Verification: Check temperature again after corrective action and confirm that it returns toward the normal operating range.

The final prediction workflow therefore functions as an end-to-end prototype. During development, the prediction script was expanded progressively from a basic fault label to the final output containing confidence, severity, action, and verification.

## 17. Confidence Interpretation

A predicted-class probability of 1.00 results in a reported confidence of 100%. **This does not mean the real-world correctness probability is 1.00 — the model probability is conditional on the learned statistical structure of the training data, and because that data is entirely simulated, high model confidence cannot establish real-world correctness.** This distinction should be stated explicitly anywhere the confidence number is presented.

## 18. Engineering Decision-Support Layer

The implemented decision-support layer performs:

**Prediction → Confidence → Severity → Recommended Action → Verification.**

## 19. Discussion

Both evaluated models achieved close to 90% aggregate accuracy, and Logistic Regression achieved the strongest overall performance despite being the simpler of the two models , added model complexity did not automatically translate into better results here. The Random Forest feature-importance analysis suggests vibration, pressure, and temperature contribute most strongly to classification, while the confusion matrix shows performance is not uniform across classes, particularly for Normal and Electrical Instability.

## 20. Technical Interpretation

- Multi-sensor fusion — the classifier uses five variables together rather than a single measurement.
- Fault classification — mapping a 5-dimensional sensor vector to one of five predefined conditions.
- Model comparison — two different algorithms evaluated on the same metrics.
- Exploratory analysis — sensor relationships examined before interpreting model results.
- Error analysis — confusion matrix used to identify class-level weaknesses.
- Decision support — classification output connected to severity, action, and verification.

- A reproducible, staged computational workflow — separate scripts for generation, training, prediction, testing, and model comparison — makes the project easier to rerun and extend.

**These characteristics place the project beyond a simple "train a classifier and print accuracy" exercise, and this claim is directly supported by the evidence reviewed.**

## 21. Limitations

### 21.1 Simulated Data

The largest limitation is that all sensor data is simulated. The model learns the statistical structure created by the simulation, not verified physical machine behaviour.

### 21.2 No Physical Sensor Validation

The project has not been validated against physical industrial sensors. Claims about real-world machine monitoring performance would be premature.

### 21.3 Limited Fault Taxonomy

Only five predefined conditions are considered; real machines can exhibit combined faults, gradual degradation, or entirely unseen conditions.

### 21.4 Lack of Temporal Modelling

The current formulation treats observations as independent feature vectors rather than a time series, even though the raw generated data includes a timestamp per row — meaning temporal information exists in the raw data but does not appear to be used as a modelling input, only (at most) as a leakage risk.

### 21.5 Generalization

The 90.60% figure describes performance on this evaluated test set only. It should not be generalized to different machines, operating loads, sensor hardware, or unseen fault types.

### 21.6 Confidence Calibration

Reported confidence values are raw model probabilities rather than independently calibrated real-world confidence. Calibration methods were not part of the current implementation and remain a possible future improvement.

### 21.7 Decision Logic Validation

The severity/action/verification layer is prototype logic, most likely rule-based, and has not been validated against real maintenance procedures.

## 22. What the Current Work Actually Represents

EdgeReliant-AI is best classified as an engineering machine-learning prototype with research-oriented experimental structure and clear potential for further research.

The project is more substantial than a basic classifier exercise because it combines multi-sensor simulation, multiple fault classes, exploratory analysis, model comparison, multiple evaluation metrics, confusion-matrix analysis, feature-importance analysis, representative multi-fault testing, confidence estimation, and a downstream engineering decision-support layer.

It should not currently be described as a completed research study, because it does not yet establish a formal research hypothesis tested through controlled experiments. The current project is the starting point for research, not a claim that research has already been completed.

## 23. Research-Oriented Potential

None of the following have been performed; they are realistic, evidence-motivated next steps.

### 23.1 Resolve the Timestamp Question First

Confirm whether timestamp was excluded from the training features. This gates the validity of every accuracy figure in this report.

### 23.2 Sensor Ablation

Remove one sensor at a time and measure the resulting change in accuracy, macro F1, and per-class recall. Research question: how does removing individual sensor modalities affect fault-classification performance?

### 23.3 Sensor Combination Experiments

Compare subsets, temperature only, vibration only, electrical sensors only, all sensors — to test whether multimodal sensing provides a measurable advantage over single modalities.

### 23.4 Noise Robustness

Inject controlled noise (5%, 10%, 20%) into sensor readings and measure degradation. Particularly relevant to eventual embedded/edge deployment credibility.

### 23.5 Cross-Validation

Replace the single 80/20 split with k-fold cross-validation, reporting mean ± standard deviation per metric.

### 23.6 Error Analysis

Investigate the Normal-class confusion identified in Sections 13-14: examine misclassified samples' raw sensor values, predicted-probability distributions, and decision boundaries.

### 23.7 Temporal Modelling

Move from single-observation classification to sequence-based classification, using engineered temporal features or sequence models. Higher effort , best scoped as a separate future project.

### 23.8 Explainability

Add SHAP or permutation importance for per-prediction explanations, building on the existing Random Forest importance ranking.

## 23.9 Edge Deployment

Measure inference latency, model size, memory usage, and CPU utilization, directly relevant to the "edge intelligence" positioning of the project and requires no new ML work.

## 23.10 Real Sensor Validation

Replace or augment simulated data with physical sensor measurements to study the simulation-to-reality domain gap. Highest value, highest effort, requires hardware access.

# 24. Proposed Research Questions for Future Work

**RQ1.** How does the combination of multiple sensor modalities affect machine fault classification compared with individual sensor measurements?

**RQ2.** How robust are machine-learning fault classifiers to increasing levels of simulated sensor noise?

**RQ3.** Which sensor modalities contribute most significantly to the classification of different fault conditions?

**RQ4.** Can temporal information improve classification of machine conditions compared with independent sample-based classification?

**RQ5.** How effectively can models trained on simulated sensor distributions generalize to physical sensor measurements?

# 25. Proposed Experimental Framework

**Dataset → Baseline → Ablation → Noise → Cross-Validation → Error Analysis → Temporal Model → Real Sensor Validation.**

The current Logistic Regression model provides the baseline for future controlled experiments. Each subsequent experiment should change one factor at a time and retain a consistent evaluation framework.

# 26. Reproducibility

The workflow is divided into separate computational stages for dataset generation, model training, prediction, multi-fault testing, and model comparison. Generated outputs and model artifacts are saved to disk, supporting repeatable execution.

# 27. Engineering Significance

The primary significance of EdgeReliant-AI is not the absolute numerical performance of the current classifier, but the demonstrated end-to-end concept:

**Physical Variables → Data → Machine Learning → Fault Diagnosis → Engineering Decision Support.**

A machine-learning model produces a prediction; an engineering system must determine what that prediction means operationally. EdgeReliant-AI attempts to bridge that gap through its severity, recommended-action, and verification layers, while keeping the decision logic separate from the learned classifier.

## 28. Conclusion

EdgeReliant-AI presents a **functional prototype** for machine-learning-based classification of simulated multi-sensor machine conditions. The project uses five sensor variables and five predefined fault states and evaluates Logistic Regression against Random Forest on a balanced dataset of 5,000 observations.

Logistic Regression achieved the strongest reported performance: Accuracy 90.60%, Macro Precision 91.82%, Macro Recall 90.60%, Macro F1 90.89%. Random Forest achieved: Accuracy 89.50%, Macro Precision 90.41%, Macro Recall 89.50%, Macro F1 89.74%. These numbers are well-supported by repeated, independent evidence and can be reported as accurate descriptions of what the code produced.

Beyond classification, the project demonstrates a genuine engineering decision-support workflow connecting predicted faults to confidence, severity, recommended action, and verification guidance. The results remain limited to the simulated environment and do not establish industrial reliability or deployment readiness.

The most appropriate description of the work is a machine-learning engineering prototype developed with an experimental and analytical mindset, providing a genuine foundation for future research into robust multi-sensor fault diagnosis and edge intelligence, not a completed research study.

## 29. Final Project Classification

**Current classification:** Engineering Machine-Learning Prototype

**Research orientation:** Strong potential

Current status as formal research: Not yet a formal research study

Primary strength: End-to-end integration of simulated sensing, ML classification, model evaluation, and engineering decision support.

**Primary limitation:** Simulation-based validation without physical sensor data, together with the need to verify the timestamp/data-leakage question before treating the reported accuracy as fully validated.

**Most valuable next step:** Verify the training feature set first, then run controlled experiments on sensor contribution and robustness, followed by real-sensor validation when hardware becomes available.